UNIVERSITY OF CENTRAL PUNJAB

Faculty of Information Technology

Cyber Security

Week 2 — Computer System Threats

Threats & Attackers • Internal vs External • Attack Vectors
Malware • Social Engineering • Network & Web Attacks

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BSCS • 3 Credit Hours • Fall 2026

QUICK RECAP

What We Learned in Week 1

Cybersecurity	Protecting our computers, networks and data — like locking our digital life.
The CIA Triad	Confidentiality, Integrity, Availability — the three goals of all security.
Key terms	Asset, threat, vulnerability, risk, exploit, attack, attack vector, payload.
This week	We zoom into THREATS — who attacks, from where, and how they get in.

WEEK 2 ROADMAP

What We Will Cover This Week

1

Computer System Threats

What a threat is, threat actors, and why systems get attacked

2

Internal vs External Threats

Dangers from inside the organization vs from outside

3

Attack Vectors

The paths and doorways attackers use to get in

4

Common Cyber Threats

Malware, social engineering, network and web attacks

What is a Computer System Threat?

A threat is anything that can harm a computer system — its data, hardware, software, or the people who use it.

In simple words: a threat is any possible danger. It becomes an actual attack only when someone (or something) acts on it.

Human threats

People who cause harm — hackers, thieves, or careless staff who click bad links

Technical threats

Malware and software problems — viruses, worms, ransomware, spyware

Natural / physical

Fire, floods, power cuts or hardware failure that destroy or stop systems

WHO ATTACKS?

Threat Actors — The People Behind Attacks

Black-hat hackers

Criminals who break in illegally for money, data, or damage.

Script kiddies

Beginners using ready-made tools they found online, often just to show off.

Insiders

Employees who misuse their access — on purpose or by mistake.

Hacktivists

Individuals or groups who use hacking and other cyber activities to promote political, social, or ideological causes

Organized crime

Professional gangs running ransomware and fraud as a business.

Nation-states

Government-backed teams that spy on or sabotage other countries.

MOTIVATION

Why Do Attackers Target Systems?

Money

Stealing bank details, running ransomware, or selling data for profit.

Information

Stealing secrets — company plans, research, or personal records.

Revenge / grudge

An angry ex-employee or rival wanting to cause damage.

Fame / challenge

Proving skills or bragging rights within hacker communities.

Politics / ideology

Pushing a cause, protesting, or attacking an opponent.

Espionage/war

Governments spying on or disrupting other nations.

TWO SOURCES OF DANGER

Internal vs External Threats

INTERNAL

Comes from INSIDE the organization

- Employees, contractors, or partners
- Already have access and trust
- May be malicious (angry staff) or accidental (a wrong click)
- Harder to detect — they look normal

EXTERNAL

Comes from OUTSIDE the organization

- Hackers, criminals, competitors
- Have no authorized access — must break in
- Use malware, phishing, and network attacks
- Blocked mainly by firewalls and strong logins

INTERNAL THREATS

Dangers From Inside

These come from people who already have access. Trust makes them dangerous.

Malicious insider

A trusted person who harms the system on purpose.

Example: An angry employee copies customer data before resigning.

Negligent insider

Someone who causes harm by carelessness, not intent.

Example: A staff member clicks a phishing link or loses a work laptop.

Weak practices

Poor habits that open the door for attackers.

Example: Sharing passwords, or leaving a computer logged in and unattended.

Defense: least privilege (only the access people need), activity logging, staff training, and quick removal of access when someone leaves.

EXTERNAL THREATS

Dangers From Outside

These come from strangers who must first break through your defenses.

Hackers

Outsiders who try to break into systems without permission.

Example: Guessing weak passwords to log into a company server.

Malware attacks

Harmful software sent from outside to infect systems.

Example: A virus hidden in a downloaded file or email attachment.

Network attacks

Attacks aimed at your internet connection and servers.

Example: Flooding a website with traffic so it crashes (DDoS).

Defense: firewalls, antivirus, software updates, strong passwords with MFA, and blocking suspicious traffic.

What is an Attack Vector?

An attack vector is the path or method an attacker uses to reach and enter a system.

Think of it like a burglar: a house can be entered through the door, a window, or the roof. Each of these is a "vector." In cyber, the doors are emails, websites, USBs, and weak passwords.

The attack surface = all the possible vectors added together. The bigger it is, the more ways in.

- **Email & phishing**
- **Malicious websites**
- **USB drives**
- **Weak passwords**
- **Unpatched software**
- **Public Wi-Fi**

ATTACK VECTORS

Common Doors Attackers Use

Phishing email

A fake email that tricks you into clicking or sharing info.

Example: "Your account is locked, click here" — leading to a fake login page.

Infected files

Malware hidden inside downloads or attachments.

Example: A free "cracked" game that secretly installs spyware.

Removable media

USBs or drives carrying malware between computers.

Example: A found USB stick that runs a virus the moment it is plugged in.

Weak / stolen logins

Guessed, reused, or leaked passwords.

Example: Using "123456" or a password already leaked in another breach.

THE MAIN CATEGORIES

Common Cyber Threats — Overview

We will look at four big families of threats in the next slides:

Malware

Harmful software: viruses, worms, trojans, ransomware, spyware

Social engineering

Tricking people: phishing, pretexting, baiting

Network attacks

Hitting connections: DoS/DDoS, man-in-the-middle, sniffing

Web & password attacks

Hitting websites & logins: SQL injection, XSS, brute force

What is Malware?

Malware (malicious software) is any program written to damage, disrupt, or secretly control a computer.

How it spreads: email attachments, downloads, infected websites, USB drives, and fake apps. It often hides inside something that looks safe.

In the next two slides we'll meet the most common types of malware.

Damages

Deletes or corrupts your files and programs

Steals

Secretly copies passwords, data and keystrokes

Controls

Takes over your device for the attacker

Types of Malware (1 of 2)

Virus	**Attaches to a file and spreads when that file is opened.** **Example:** An infected document that infects others once you open it.
Worm	**Copies itself across networks with no human help.** **Example:** Spreads computer-to-computer on its own, clogging the network.
Trojan	**Looks like a useful app but hides harm inside.** **Example:** A "free PDF reader" that secretly opens a backdoor.
Spyware	**Secretly watches and reports what you do.** **Example:** Records your keystrokes to steal passwords (keylogger).

Types of Malware (2 of 2)

Ransomware	**Locks your files and demands payment to unlock them.** **Example:** WannaCry encrypted files worldwide and demanded Bitcoin.
Adware	**Floods you with unwanted ads and pop-ups.** **Example:** Endless pop-ups after installing a free program.
Rootkit	**Hides deep in the system to avoid being found.** **Example:** Gives an attacker hidden admin control of your PC.
Botnet	**Many infected devices controlled together by an attacker.** **Example:** Thousands of hijacked PCs used to attack a website at once.

Hacking the Human, Not the Machine

Social engineering means tricking people into giving away information or access — instead of breaking the technology.

Why it works: it targets human emotions — trust, fear, curiosity, or urgency. It's often easier to fool a person than to defeat a firewall.

Fear

"Your account will be closed!"

Urgency

"Act in the next 10 minutes!"

Curiosity

"See who viewed your profile"

Trust

Pretending to be your boss or bank

Common Social Engineering Attacks

Phishing	**Mass fake emails/texts to trick many people at once.** **Example:** "Your bank needs you to verify your password here."
Spear phishing	**A targeted, personalized version aimed at one person.** **Example:** An email using your real name and job title to seem real.
Pretexting	**Inventing a believable story to gain trust.** **Example:** A caller pretending to be IT support asking for your password.
Baiting	**Offering something tempting to lure a victim.** **Example:** A "free movie download" or a USB left in the car park.

Network-Based Attacks

DoS (Denial of Service)

Definition: Makes a system/server/network unavailable by sending more requests or traffic than it can handle.

Example: Thousands of requests from one computer make a university website slow or unavailable.

DDoS (Distributed Denial of Service)

Definition: Similar to DoS, but traffic comes from many compromised devices at the same time.

Example: Thousands of infected computers send traffic to an online banking server.

Man-in-the-Middle (MITM)

Definition: An attacker secretly places themselves between two communicating parties and may monitor or modify communication.

Example: An attacker on insecure public Wi-Fi intercepts communication between a student and a website.

Packet Sniffing

Definition: Captures and examines network packets traveling between devices; it can be legitimate for troubleshooting or malicious.

Example: An attacker captures traffic on an insecure network and examines it for usernames or other information.

Credential Sniffing

Definition: Captures authentication information such as usernames and passwords from network traffic or other communication channels.

Example: A login over an insecure connection is captured by an attacker.

Web & Authentication Attacks

SQL Injection (SQLi)
Definition: Inserts malicious SQL commands into application input to manipulate a database.
Example: A vulnerable login form accepts database commands through the username field.

Cross-Site Scripting (XSS)
Definition: Injects malicious script, usually JavaScript, into a trusted website so it executes in another user's browser.
Example: A malicious script is placed in a comment and runs when another user views it.

Brute-Force Attack
Definition: Repeatedly tries different passwords or authentication combinations until the correct one is found.
Example: An attacker repeatedly tries different passwords against a login account.

Session Hijacking
Definition: Obtains or takes control of a user's active session, often by stealing a session token.
Example: An attacker obtains a user's session token and uses it to access the account.

Session Fixation
Definition: Causes a victim to use a session identifier that the attacker already knows.
Example: The victim logs in using a session ID previously supplied by an attacker.

Spoofing & Social Engineering

Spoofing
Definition: Pretends to be a trusted device, user, website, or service by falsifying identifying information.
Example: An email appears to come from the university IT department.

IP Spoofing
Definition: Changes the source IP address of packets to make them appear to come from another device.
Example: Packets use a forged source IP to hide the true origin of traffic.

ARP Spoofing
Definition: Sends fake ARP messages to associate the attacker's device with another device's IP address and redirect traffic.
Example: A computer is tricked into believing the attacker's machine is the network gateway.

DNS Spoofing
Definition: Provides false DNS information so users are redirected to an incorrect or malicious destination.
Example: A legitimate website address redirects the user to a fake website.

DNS Poisoning
Definition: Inserts incorrect information into a DNS cache, causing users to receive the wrong IP address.
Example: A poisoned cache redirects requests for a bank website to a fraudulent server.

Phishing
Definition: Tricks users into revealing sensitive information or clicking a malicious link.
Example: A fake university email asks a student to verify an account through a fake login page.

Wireless, Network & Reconnaissance Attacks

Evil Twin Attack

Definition: Creates a fake Wi-Fi access point that looks like a legitimate network.

Example: An attacker creates 'University_Free_WiFi' and waits for students to connect.

Port Scanning

Definition: Checks a computer/server for open network ports and services; used legitimately for testing or by attackers for reconnaissance.

Example: An administrator scans a server to identify exposed services.

Replay Attack

Definition: Captures valid network communication and later retransmits it to trick a system into accepting it again.

Example: A captured authentication message is retransmitted to attempt access.

DHCP Starvation

Definition: Consumes available DHCP addresses by sending many requests, preventing legitimate devices from obtaining IP addresses.

Example: Fake DHCP requests exhaust the server's available addresses.

DHCP Spoofing

Definition: An unauthorized DHCP server provides false network configuration information to clients.

Example: A fake DHCP server tells users to use the attacker's machine as their gateway.

MAC Spoofing

Definition: Changes a device's MAC address to impersonate another device or bypass network controls.

Example: An attacker changes their MAC address to match an authorized computer.

Wi-Fi Deauthentication Attack

Definition: Sends forged wireless management frames to disconnect devices from a Wi-Fi network.

Example: A student's laptop is repeatedly disconnected from campus Wi-Fi.

Advanced & Browser-Based Attacks

Man-in-the-Browser (MITB)

Definition: Uses malicious software inside a web browser to monitor or modify transactions on a legitimate website.
Example: A Rs. 5,000 bank transfer is secretly changed by malware before reaching the bank.

Key Concepts to Remember

Definition: Attacks can target availability, confidentiality, integrity, authentication, or user trust.
Example: DoS/DDoS → availability; Sniffing/MITM → confidentiality; SQLi/XSS → application security; Spoofing/Phishing → trust and identity.

Security Perspective

Definition: The same technique may be used legitimately for security testing or maliciously by attackers.
Example: Port scanning can help an administrator identify exposed services, but attackers may use it for reconnaissance.

CONSEQUENCES

What Damage Can Threats Cause?

Data loss / theft

Personal, financial or business data is stolen or destroyed.

Financial loss

Ransom payments, fraud, downtime, and recovery costs.

Reputation damage

Customers lose trust; the brand's name is harmed.

Service downtime

Websites and systems go offline, stopping work.

Legal penalties

Fines and lawsuits for failing to protect user data.

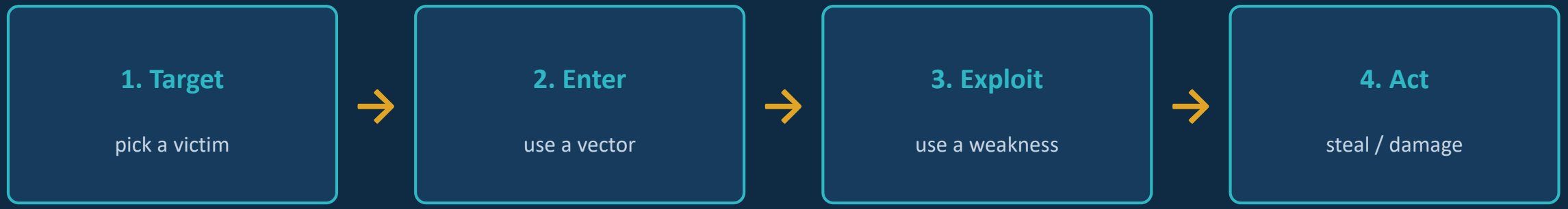
Privacy invasion

Private messages, photos, or identities are exposed.

PUTTING IT TOGETHER

How a Typical Attack Unfolds

Most attacks follow a simple pattern — read it left to right:



Real-life example

An attacker picks a company (target), **sends a staff member a phishing email** (vector). The employee clicks and installs malware that uses an unpatched bug (exploit), then the attacker steals customer data **(act).**

STAYING SAFE

Basic Defenses Against Threats

Strong passwords + MFA

Long, unique passwords plus a second step to log in.

Update software

Patches close the weaknesses attackers exploit.

Antivirus & firewall

Detect malware and block unwanted traffic.

Think before you click

Don't open unknown links, files, or USBs.

Back up data

Keep copies so ransomware can't hold you hostage.

Least privilege

Give people only the access they truly need.

RECAP

Week 2 Summary

- A threat is anything that can harm a system; a threat actor is the person or group behind it.
- Threats are internal (from trusted insiders) or external (from outside attackers).
- An attack vector is the path in — email, files, USBs, weak passwords, unpatched software.
- Main threat families: malware, social engineering, network attacks, and web/password attacks.
- Malware types include virus, worm, trojan, spyware, ransomware, adware, rootkit and botnet.
- Basic defenses: strong passwords + MFA, updates, antivirus, backups, and careful clicking.

CHECK YOUR UNDERSTANDING

Review Questions

1. What is a computer system threat? Name three types of threat actors.
2. Explain the difference between internal and external threats, with an example of each.
3. What is an attack vector? List four common attack vectors.
4. Describe four types of malware in your own words.
5. What is social engineering, and why does it work on people?
6. Give two basic defenses against cyber threats and explain how each helps.

WEEK 3 PREVIEW

Coming Next: Vulnerabilities & CVE

- Definition of vulnerabilities
- Common system vulnerabilities
- Introduction to CVE (Common Vulnerabilities and Exposures)